Series and Sequences Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 An arithmetic sequence has the first term 5 and a common difference of d. The sum of the first 20 terms is 4 times the sum of the first 10 terms. Find the value of d. 2

$$S_{20} = 4S_{10}$$
$$\frac{20}{2}\left[2(5)+19d\right] = 4\times\frac{10}{2}\left[2(5)+9d\right]$$
$$10(10+19d) = 20(10+9d)$$
$$10+19d = 20+18d \qquad d=10$$

2 Four times the sum of the first two terms of an infinite geometric progression is equal to the limiting sum. Determine the possible values of the common ratio. 2

$$\frac{a}{1-r} = 4(a+ar) \qquad a = 4a(1-r^2)$$
$$\frac{1}{4} = 1-r^2 \qquad r^2 = \frac{3}{4} \qquad r = \pm\frac{\sqrt{3}}{2}$$

3 Consider the geometric series $1 + (\sqrt{5}-2) + (\sqrt{5}-2)^2 + \cdots$

(i) Show with calculations why this geometric series has a limiting sum. 1

Limiting sum when $|r|<1$

$a=1$, $r=\sqrt{5}-2$ $\quad |r| = |\sqrt{5}-2| \doteqdot 0.236 < 1$

$\therefore$ limiting sum exists.

(ii) Find the exact value of the limiting sum with a rational denominator. 2

$$S = \frac{a}{1-r} = \frac{1}{1-(\sqrt{5}-2)} = \frac{1}{3-\sqrt{5}}$$
$$= \frac{1}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}} = \frac{3+\sqrt{5}}{4}$$

4 A city is suffering through a drought and adopts a plan to import water from another city. It is given that the volume of water imported in the 1st year since the start of the plan is $1.5\times10^7\ m^3$ and in subsequent years, the volume of water imported each year is 10% less than the volume of water imported in the previous year.

1st Yr $\quad$ 2nd Yr $\quad$ 3rd Yr

$1.5\times10^7 \xrightarrow{\times 90\%} 1.35\times10^7 \xrightarrow{\times 90\%} 1.215\times10^7 \ \ldots$

$a = 1.5\times10^7 \quad r = 0.90$

(i) Find how much water was imported in the 8th year. 1

$T_8 = a.r^{n-1}$

$= 1.5\times10^7 \times (0.9)^{8-1}$

$\doteqdot 7.17\times10^6\ m^3$

(ii) Find the total volume of water imported in the first 10 years. 1

$$S_{10} = \frac{1.5\times10^7\left[0.9^{10}-1\right]}{0.9-1}$$
$\doteqdot 9.8\times10^7\ m^3$

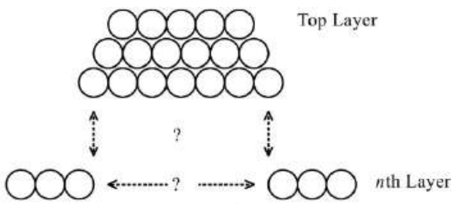
(iii) Explain why the total volume of water imported since the start of the plan will never exceed $1.6\times10^8\ m^3$. 2

$$S_\infty = \frac{a}{1-r} = \frac{1.5\times10^7}{1-0.9} = \frac{1.5\times10^7}{0.1}$$
$$= 1.5\times10^8$$

$1.5\times10^8 < 1.6\times10^8$

$\therefore$ By limiting sum can never exceed $1.6\times10^8\ m^3$

5 Lachlan works in a Woolworths store, and he is making a stack of oranges against a sloping display panel. The oranges are stacked in layers, as shown, where each layer contains one orange less than the layer below it. When he has finished, there are five oranges in the top layer, six in the next and so on. There are n layers altogether.



(i) Show that there are $\frac{1}{2}n(n+9)$ oranges in the sack. 2

$a = 5 \quad d = 1$

$S_n = \frac{n}{2}[2(5) + (n-1) \times 1]$

$= \frac{n}{2}[10 + n - 1]$

$= \frac{n}{2}(n+9) = \frac{1}{2}n(n+9)$

(ii) If Lachlan has 300 oranges to create his display, how many full rows can he create, if the top row still contains five oranges? 2

$300 = \frac{1}{2}n(n+9)$

$600 = n^2 + 9n$

$0 = n^2 + 9n - 600$

$n = \frac{-9 \pm \sqrt{9^2 - 4(1)(-600)}}{2} = \boxed{20.4}08 \quad -29.4$

$\therefore$ 20 Rows can be created.

6 In the first week of the snow season 5 cm of snow falls. In each of the following weeks the snowfalls increase by 2 cm, so that in the second week there is a 7 cm snow fall, in the third week there is 9 cm. This continues to the middle week and, from then on, weekly snowfall decreases by 2 cm per week, the season lasting 21 weeks.

(i) How much snow falls in the 11th week? 1

AP $5, 7, 9, 11 \ldots \quad a = 5 \quad d = 2 \quad n = 11$

$T_n = a + (n-1)d$

$T_{11} = 5 + 10 \times 2 = 25$ cm.

(ii) What is the total snowfall for the whole season? 3

Snowfall $= \cancel{2+}\; \overset{T_1}{5} + \overset{T_2}{7} + \overset{T_3}{9} + \ldots + \overset{T_{11}}{25} + \overset{T_{12}}{23} + \ldots \overset{T_{20}}{7} + \overset{T_{21}}{5}$

$= 2 \times S_{10} + 25$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$S_{10} = \frac{10}{2}[10 + 9 \times 2] = 5 \times 28 = 140$ cm

Total snowfall $= 2 \times 140 + 25 = 305$ cm

7 For $m \neq 0$, what is the limiting sum of the geometric series $m + \frac{m}{1+m^2} + \frac{m}{(1+m^2)^2} + \frac{m}{(1+m^2)^3} + \cdots$? 2

$a = m \qquad r = \frac{1}{1+m}$

$$S_\infty = \frac{m}{1 - \frac{1}{1+m^2}} = \frac{m \times (1+m^2)}{\left(1 - \frac{1}{1+m^2}\right) \times (1+m^2)} = \frac{m+m^3}{1+m^2-1}$$

$$= \frac{m+m^3}{m^2} = \frac{1+m^2}{m}$$

8 Jane thinks about three distinct numbers $x, x+4$ and $9x$. Find all possible sets of these three numbers if they are successive terms of a geometric series. 3

$$\frac{x+4}{x} = \frac{9x}{x+4}$$

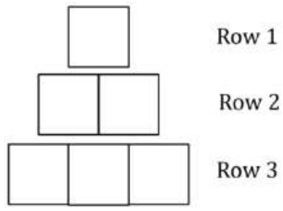
$(x+4)^2 = 9x^2 \qquad x+4 = \pm 3x$

$x+4 = -3x \qquad x+4 = 3x$

$x = -1 \qquad x = 2$

$\therefore$ Series could be $-1, 3, -9$ or $2, 6, 18$

9 James has some sticks that are all of the same length. He arranges them in squares and has made the following 3 rows of patterns:



He notices that 4 sticks are required to make the single square in the first row, 7 sticks to make 2 squares in the second row and in the third row he needs 10 sticks to make 3 squares.

(i) Find an expression, in terms of n, for the number of sticks required to make a similar arrangement of n squares in the nth row. 2

$\{4, 7, 10, \ldots\}$
Arithmetic series with first term 4 and common difference 3.
$T_n = a + (n-1)d = 4 + 3(n-1) = 3n + 1$

(ii) James continues to make squares following the same pattern. He makes 4 squares in the 4th row and so on until he has completed 10 rows. Find the total number of sticks James uses in making these 10 rows. 1

$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{10}{2}[2 \times 4 + (10-1) \times 3] = 175$

(iii) James started with 1750 sticks. Given that James continues the pattern to complete k rows but does not have sufficient sticks to complete the (k+1)th row, show that k satisfies $(3k - 100)(k + 35) < 0$. 2

The number sticks is less than 1750. $S_k < 1750$

$$\frac{k}{2}[2 \times 4 + 3(k-1)] < 1750$$
$$k(8 + 3k - 3) < 3500$$
$$3k^2 + 5k - 3500 < 0$$
$$\therefore (3k - 100)(k + 35) < 0$$

(iv) What is the value of k? 1

$(3k - 100)(k + 35) = 0$
$k = \frac{100}{3}$ or $k = -35$
The value of k is a positive whole number (rows).
$\therefore k = 33$

Series and Sequences Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Which expression is a term of the geometric series $2x - 6x^3 + 18x^5 - \cdots$? 1

A. $1458x^{13}$ B. $-1458x^{13}$

C. $1458x^{15}$ D. $-1458x^{15}$

G.P. $a = 2x$, $r = -\frac{6x^3}{2x} = -3x^2$

$T_n = a \times r^{n-1}$

$T_7 = 2x \times (-3x^2)^{7-1}$

$= 2x \times 729x^{12}$

$= 1458x^{13}$ $\therefore \boxed{A}$

2 The first term of an arithmetic series is 4 and the third term is 10. What is the 20th term of the series? 2

$a = 4$

$T_3 = 10$

$a + (3-1)d = 10$

$4 + 2d = 10$

$d = 3$

$\therefore T_{20} = 4 + (20-1) \times 3 = 61$

3 The fourth term of an arithmetic series is 27 and the seventh term is 12. What is the common difference? 1

$T_4: a + 3d = 27$ ①

$T_7: a + 6d = 12$ ②

$3d = -15$ ② − ①

$d = -5$

4 The first term of an arithmetic sequence is 5. The sum of the first 43 terms is 2021. What is the common difference of the sequence? 2

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2021 = \frac{43}{2}(2 \times 5 + (43-1)d)$$

$$94 = 10 + 42d$$

$$d = \frac{94 - 10}{42} = 2$$

5 Solve $1 + \left(\frac{x+1}{x-1}\right) + \left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x+1}{x-1}\right)^3 + \cdots = \frac{x^2}{2}$. 3

Limiting sum: $S_\infty = \frac{a}{1-r}$, $a = 1$, $r = \frac{x+1}{x-1}$

$\frac{1}{1 - \frac{x+1}{x-1}} = \frac{x^2}{2}$

$\frac{x-1}{x-1-x-1} = \frac{x^2}{2}$

$\frac{x-1}{-\cancel{2}} = \frac{x^2}{\cancel{2}}$

$\therefore x^2 + x - 1 = 0$

$x = \frac{-1 \pm \sqrt{5}}{2}$

but for limiting sum to exist, $-1 < r < 1$

sub $x = \frac{-1+\sqrt{5}}{2}$ into r: $r \approx -4.23$

sub $x = \frac{-1-\sqrt{5}}{2}$ into r: $r \approx 0.23$

$\therefore x = \frac{-1-\sqrt{5}}{2}$

6 The sum of the first n terms of an arithmetic series is given by $S_n = 2n(n+2)$. Find the first term and the common difference of the series. 2

$S_n = 2n(n+2)$ $\therefore T_1 = S_1 = 6$ and

$T_2 = S_2 - S_1 = 16 - 6 = 10$

Hence common difference is $T_2 - T_1 = 4$

7 Find the values of x and y if the first four terms of a geometric sequence are 3, x, y, 192. 2

$\frac{x}{3} = \frac{y}{x} = \frac{192}{y}$

$\therefore x^2 = 3y$ — ①

$y^2 = 192x$ — ②

Subst ② into ①

$\left(\frac{y^2}{192}\right)^2 = 3y$

$\frac{y^4}{36864} = 3y$

$y = 48$

$\therefore x = 12$ & $y = 48$.

8 Suppose we have an arithmetic series with $T_6 = -10$ and $T_{21} = 50$, and an infinite geometric series with $T_1 = 32$ and common ratio $\frac{3}{4}$.

(i) Find the limiting sum of the geometric series. $S_\infty = \frac{a}{1-r} = \frac{32}{1-\frac{3}{4}} = 128$ 1

(ii) Show that the sum of the first n terms of the arithmetic series is $S_n = 2n^2 - 32n$. 3

$$T_6 = a + 5d = -10 \ldots (1)$$
$$T_{21} = a + 20d = 50 \ldots (2)$$
$$(2) - (1): \ 15d = 60$$
$$d = 4 \text{ and } a = -30.$$
$$S_n = \frac{n}{2}(2a + (n-1)d)$$
$$= \frac{n}{2}(-60 + 4(n-1))$$
$$= \frac{n}{2}(4n - 64) = 2n^2 - 32n$$

(iii) Find the value of n for which the sum of the arithmetic series is three times the limiting sum of the geometric series. 2

$$2n^2 - 32n = 384$$
$$n^2 - 16n - 192 = 0$$
$$(n-24)(n+8) = 0$$
$n = 24$ since n is positive.

9 Consider the geometric series $1 + \frac{4}{3}sin^2 x + \frac{16}{9}sin^4 x + \frac{64}{27}sin^6 x + \cdots$

(i) When the limiting sum exists, find an expression for its value. 1

$r = \frac{4}{3}\sin^2 x$ $\qquad S_\infty = \frac{1}{1-\frac{4}{3}\sin^2 x}$

$a = 1$

(ii) For what values of x in the interval $0 < x < \frac{\pi}{2}$ does the limiting sum of this series exist? 2

S_∞ exists for $|r|<1$ ie $-1<r<1$

$\therefore \left|\frac{4}{3}\sin^2 x\right| < 1$ ie $-1 < \frac{4}{3}\sin^2 x < 1$

$0 < \frac{4}{3}\sin^2 x < 1$ in Q1

$0 < \sin^2 x < \frac{3}{4}$

$0 < \sin x < \frac{\sqrt{3}}{2}$

$\therefore 0 < x < \frac{\pi}{3}$

10 Tom planted a silky oak tree three years ago when it was 80 cm tall. At the end of the first year after planting, it was 130 cm tall, that is it grew 50 cm. Each year's growth was 90% of the previous year's.

(i) What was the growth of the silky oak after 3 years? 1

$80 \quad 50 \quad 50(0.9) \quad 50(0.9)^2 \quad 50(0.9)^3 \ldots$

year 1 year 2 year 3

Growth after 3 years $= 50 + 50(0.9) + 50(0.9)^2 = 135.5$ cm

(ii) Assuming that it maintains the present growth pattern, explain why the tree will never reach a height of 6 metres. 2

$50, 50(0.9), 50(0.9)^2, \ldots \Rightarrow$ G.P. $r = 0.9$, infinite series

S_∞ exists as $|r|<1$

$S_\infty = \frac{a}{1-r} = \frac{50}{1-0.9} = 500$ cm

$\therefore$ limiting growth is 500 cm & limiting height is 580 cm < 600 cm

$\therefore$ Tree will never reach a height of 6 m

(iii) In which year will the silky oak reach a height of 5 metres? 2

To reach a height of 5 m, tree needs to grow 420 cm

$S_n = \frac{a(1-r^n)}{1-r}$ $\qquad 0.9^n = 1 - \frac{42}{50}$

$\qquad 0.9^n = \frac{8}{50}$

$420 = \frac{50(1-0.9^n)}{1-0.9}$ $\qquad \log_{0.9}\left(\frac{4}{25}\right) = n$

$42 = 50(1-0.9^n)$ $\qquad n = \frac{\ln(0.16)}{\ln(0.9)}$

$\frac{42}{50} = 1 - 0.9^n$

$n \doteqdot 17.39344$

$\therefore$ Tree will reach 5 metres during 18th

11 On his 1st birthday, Timmy was given \$5 by his Aunty Ruth. On each subsequent birthday, Aunty Ruth gave Timmy \$2 more than the previous year.

(i) How much did Aunty Ruth give Timmy on his 20th birthday? 1

$T_n = a + (n-1)d$

$= 5 + (20-1) \times 2 = 43$

Timmy was given \$43.

(ii) If Timmy saves the money every year, how old will he be when he has received over \$100 in total? 2

$S_n = \frac{n}{2}(2a + (n-1)d) > 100$

$\frac{n}{2}(10 + (n-1)2) > 100$

$n(n+4) > 100$

By trial & error, $n > 8$

He has received over \$100 after his 9th Birthday